PROJECT REPORT

ANALYSIS OF ITCH.IO TOP SELLERS

Web scraping, data quality and exploratory data analysis

Course	DSC2107 Data Mining & Wrangling
Group	Group 8
Snapshot period recorded in notebook	13–19 October 2026
Report basis	Data collected over seven days

Group members

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Prepared from the files supplied for the Group 8 project.

Executive Summary

This report documents Group 8's web-scraping and exploratory-analysis project on itch.io's Top Sellers listing. The supplied scraper is designed to collect public listing-card fields, preserve raw CSV and JSON snapshots, create a separate cleaned CSV, and store game details and date-specific observations in SQLite. The saved outputs in Analysis.ipynb report seven daily snapshots from 13 to 19 October 2026, with 500 observations per date (3,500 observations) and 852 distinct game URLs across the week.

The notebook reports a median parsed price of 2.00, Q1 = 0.00, Q3 = 5.99, 189 IQR-defined outlier observations, and a Spearman correlation of −0.044 between observed rank and parsed price. The graph shows Adventure, Visual Novel, Simulation and Action as the most frequently recorded genre/tag labels. The output contains a flat count of 500 unique URLs per date. Platform analysis is incomplete: 84.6% of platform values are missing and the frequency graph only displays "Play in browser".

Important evidence qualification

The notebook output says it loaded seven collected CSV snapshots, but the actual seven CSV files were not included in the uploaded package. This report therefore uses the saved notebook outputs as evidence and cannot independently re-run all calculations. Also, the current price parser may treat "Free / no price displayed" as zero, so price statistics and the free share require validation before final claims.

1. Introduction

Web scraping is the automated collection of information displayed on a webpage. The selected target is the public itch.io Top Sellers listing. The project transforms visible game-card information into structured records, checks and cleans fields, stores observations, then uses charts to identify patterns. The focus is descriptive analysis of observed listings, not exact sales volumes, revenue or developer profits. [1–3]

2. Aim and Objectives

Aim: to collect, clean, store and analyze daily snapshots of itch.io Top Sellers listings through a reproducible Python workflow.

- Collect up to 500 distinct listing cards per daily run, with rank, title, developer, displayed price, genre, platforms, URL and timestamp.
- Keep daily raw CSV/JSON evidence unchanged and write cleaned data separately.
- Store stable game information and daily observations in related SQLite tables.
- Use exploratory charts to study prices, rank-price association, daily changes, developer frequency, genres and platform labels.
- Report missingness and limitations and avoid unsupported sales, revenue, ratings or discount claims.

3. Methodology and Dataset Profile

3.1 Collection workflow

- Access the target https://itch.io/games/top-sellers only after checking robots.txt; identify permitted HTML pages and use a course-identifying User-Agent.
- Request listing pages with requests.Session and parse HTML using BeautifulSoup. CSS selectors are tried for game cards, titles, developers, prices, genre labels and platform labels.
- Assign a sequential rank based on listing order, avoid duplicate URLs within a run, follow pagination, and stop at 500 distinct games or when no usable cards are found.
- Use 2–5 second request delays, a per-run request cap, bounded retries/back-off, logging, and stop-on-401/403/429 behaviour rather than bypassing access blocks.
- Save dated raw CSV and JSON files, create a separate cleaned CSV and update the SQLite products and observations tables. [1, 2]

3.2 Processing in Analysis.ipynb

The notebook loads and combines CSV snapshots, standardizes text fields, converts rank to numeric, converts timestamps to UTC datetime, parses price values, creates snapshot_date, removes rows missing title or URL, and removes duplicate combinations of game URL and timestamp. It then displays data types, missing counts, unique URL counts and observations by day. [3]

3.3 Snapshot profile and data quality

Measure	Saved notebook output	Meaning
Snapshot dates	13–19 Oct 2026 (7)	Seven consecutive dates
Observations	3,500 total; 500/day	Daily target count recorded
Distinct game URLs	852 over the period	Listings overlap across dates
Missing genre labels	337 (9.6%)	Genre chart does not cover every row
Missing platform labels	2,961 (84.6%)	Platform extraction is highly incomplete
Rank/title/developer/price text/timestamp	0 missing recorded	Complete after notebook cleaning; does not itself validate correctness

3.4 Storage model and questions

The products table stores stable game details (URL, title, developer, genre and platforms), while observations stores per-date rank and price fields, listing page and scrape timestamp. The notebook addresses seven questions: price distribution; price versus rank; seven-day changes; developers most frequently observed; free/paid share; most common genres and platforms; and price variation by genre. Ratings/reviews and verified original prices are not in the assigned fields, so those analyses are unsupported. [1–3]

4. Price Distribution and Rank Association

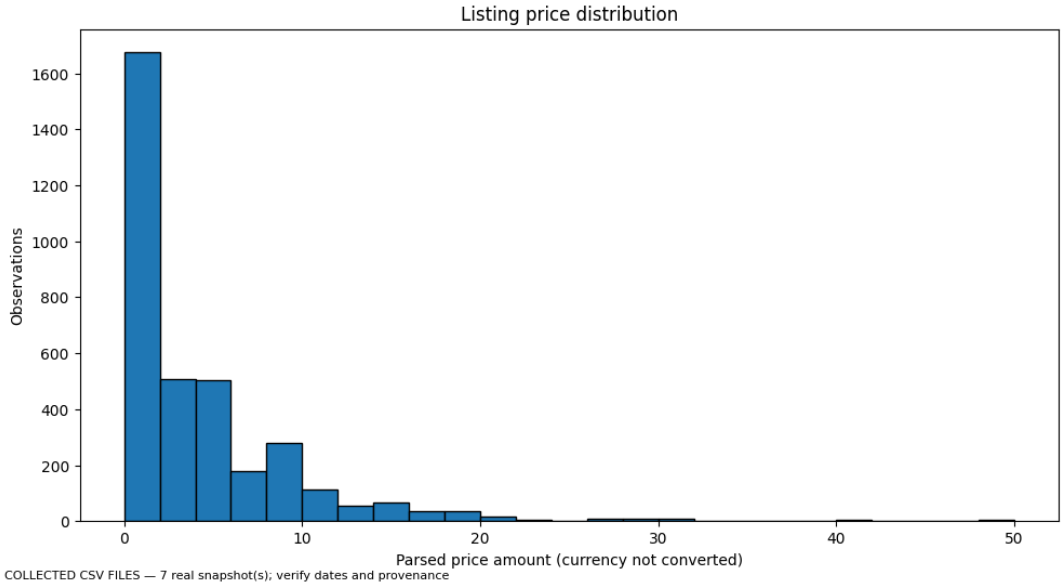


Figure 1. Listing price distribution; cropped from the saved Analysis.ipynb chart output.

The histogram is strongly right-skewed: many parsed values cluster near zero and a smaller number extend to higher amounts. The notebook reports median = 2.00, Q1 = 0.00, Q3 = 5.99 and 189 outlier observations by the IQR rule. IQR = Q3 − Q1 = 5.99. An IQR outlier is a statistical flag, not automatically an error. The x-axis states that prices were not converted to a common currency. [3]

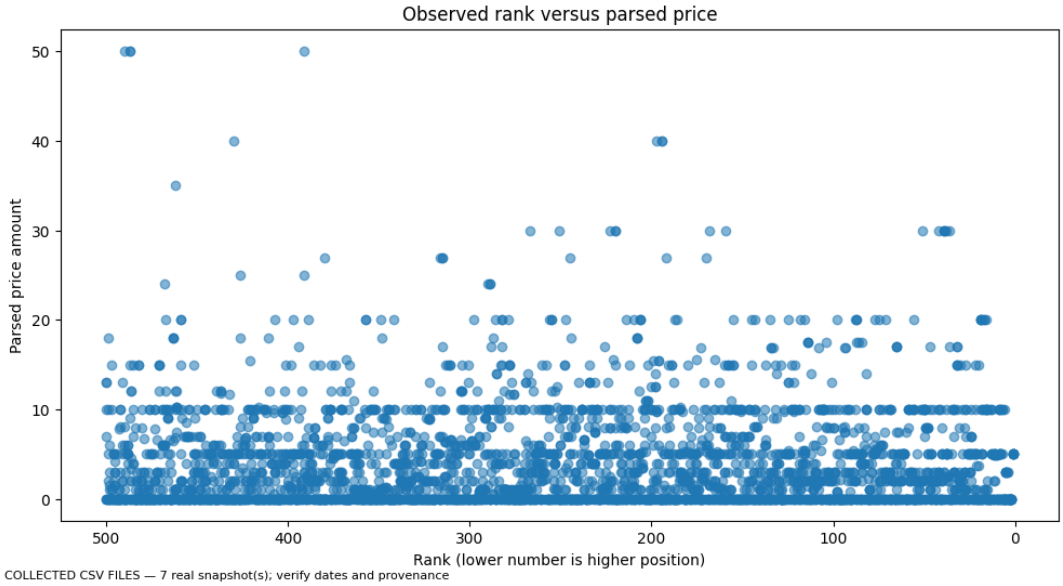


Figure 2. Observed rank versus parsed price; lower rank numbers represent higher positions.

The scatter plot displays individual observations. The saved Spearman correlation is −0.044, which is very close to zero and suggests little monotonic association between parsed price and rank in this sample. It does not show causation. Rank is the scraper-assigned listing order, not a measure of units sold, and the project does not have verified ratings or review counts. [2, 3]

5. Changes Across the Seven Snapshots

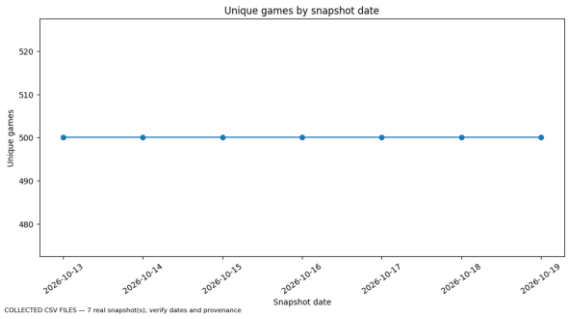


Figure 3. Unique game URLs by snapshot date.

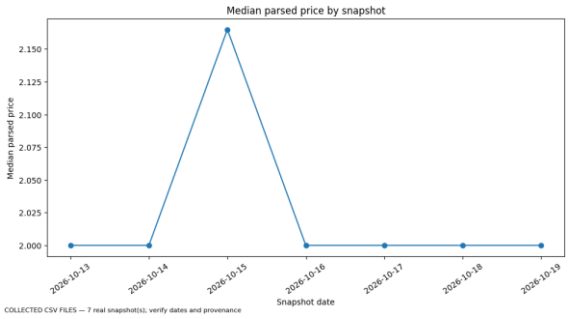


Figure 4. Median parsed price by snapshot.

5.1 Number of unique games

The saved line graph is flat at 500 unique URLs for each of the seven collection dates. This is consistent with the collection target of 500 listings per daily run. It does not mean the same 500 games were present every day; the notebook reports 852 distinct URLs across the full period. [3]

5.2 Median price over time

The median-price line is close to 2.00 on six of the seven dates, with a visible temporary peak on 15 October before returning to about 2.00. Exact daily median values were not printed in a table in the saved notebook output, so this is a visual interpretation only. The ambiguous price fallback described in Section 7 also affects confidence in these price results. [3]

5.3 Rank movement

For a game observed on at least two dates, the notebook calculates rank range = maximum observed rank − minimum observed rank. Larger values indicate more variation in recorded position. Movement can be affected by a game entering or leaving the top-500 window and does not measure changes in sales.

6. Ranking Movement and Developer Frequency

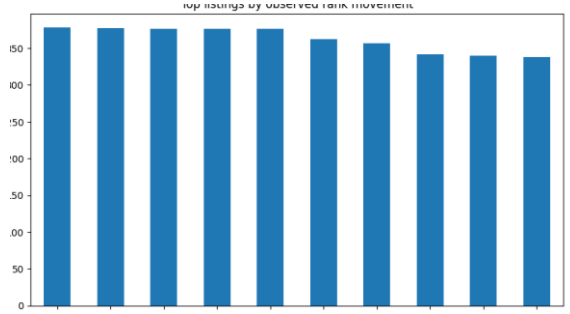


Figure 5. Cropped rank-movement plot; exact labels and rank ranges are listed below.

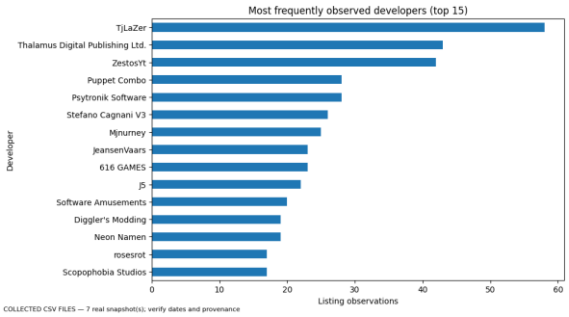


Figure 6. Developers most frequently observed in the listing records.

The largest ranges in the saved table are summarized below. Several games have missing observations on some dates, so each range uses the available ranks and should not be interpreted as uninterrupted movement. The developer chart counts recorded appearances and may count a game/developer again on later dates.

Game title	Rank range (max – min)
This Role: You are Hamster	378
Steel Legion	377
OneShot	376
Isolated Inhale	376
The Gardener	376
Project: Depot Shift	362
Bruce Lee for Atari ST	357
Dweller's Empty Path	342
ISO/2004 [Chapter#1]	340
Repugnant Bounty	338

In the top-15 developer chart, TilaZer appears most often, followed by Thalamus Digital Publishing Ltd. and ZestosYt. These are counts in this sample; they are not evidence of total game output, revenue, sales or market share. [3]

7. Genre, Platform and Price-Group Patterns

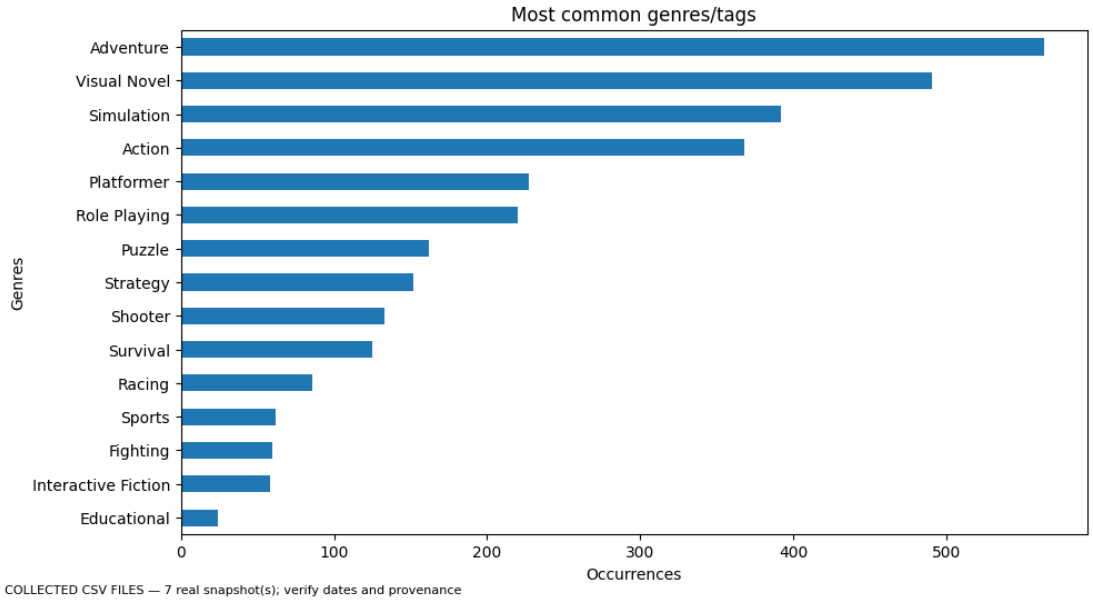


Figure 7. Most frequently recorded genre/tag labels.

Adventure is the most frequent recorded label, followed by Visual Novel, Simulation and Action. Multiple labels separated by a vertical bar are split before counting, so one record may contribute to more than one category. The chart covers available labels; 9.6% of genre values are missing. [3]

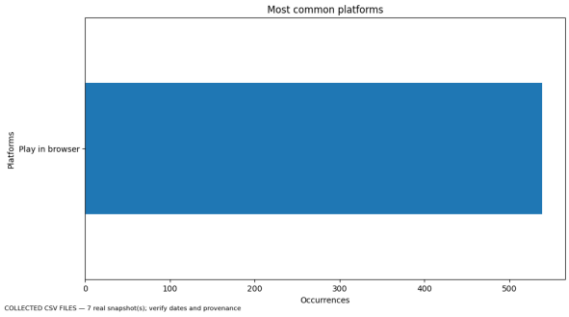


Figure 8. Platform labels observed in the saved output.



Figure 9. Parsed free versus positive-price observations.

Only “Play in browser” appears in the platform-frequency chart. Given that 84.6% of platform fields are missing, this is evidence of incomplete extraction, not proof that the games support only browser play. The saved price-group chart reports a free share of 35.9%, but that share requires correction/validation because missing-price text can be converted into numeric zero.

Price classification must be fixed before final claims

In scraper.py, a missing price element is labelled “Free / no price displayed”. The cleaner then converts any text starting with “Free” to 0.0. A missing display is not necessarily a free price. The dataset should distinguish explicit free, numeric price and unknown/not displayed, then rerun the price, median and free/paid analyses.

8. Price Variation by Genre, Discussion and Limitations

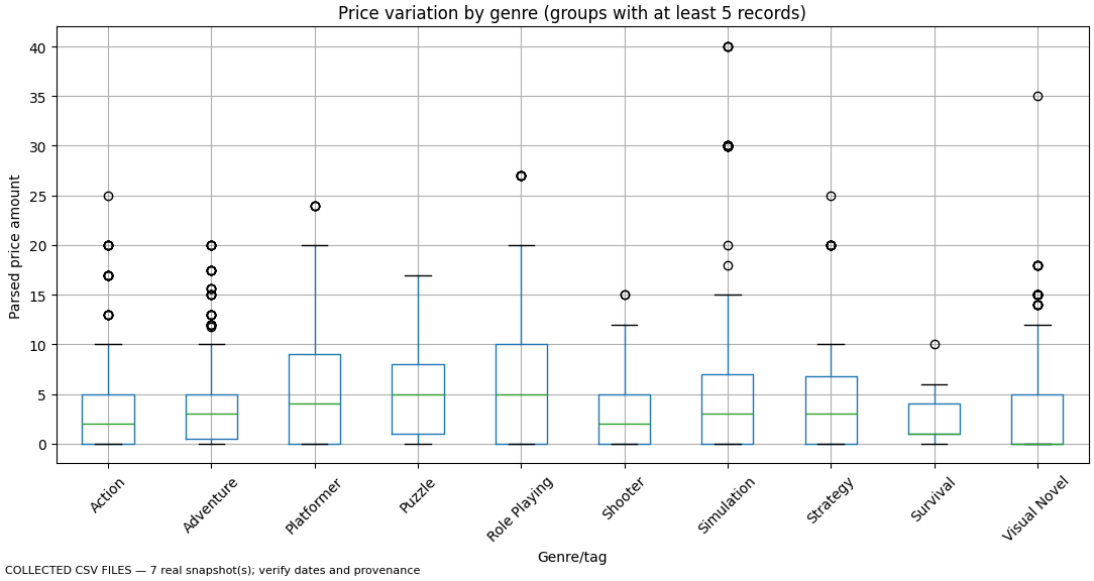


Figure 10. Price distributions by selected genre/tag; groups with at least five records.

The box plot compares the median and spread of parsed price values across selected genre/tag groups. The line inside a box is the median, the box spans the middle 50% of values, and points beyond the whiskers are potential outliers. Different groups appear to have different spreads and high-price points. These comparisons are preliminary until ambiguous price values are corrected. [3]

8.1 Main findings

- The saved notebook combines 3,500 observations from seven dates and reports 852 distinct game URLs.
- Price values are concentrated near zero, with a long upper tail; 189 records are flagged by the IQR outlier rule.
- The rank-price Spearman correlation is −0.044, indicating little monotonic association in the observed sample.
- Adventure, Visual Novel, Simulation and Action are the most frequent available genre/tag labels.
- High platform missingness prevents a reliable conclusion about supported platforms.

8.2 Key limitations

- The source CSV snapshots were not uploaded with the notebook, so the report cannot independently reproduce every metric.
- The “Free / no price displayed” fallback may be incorrectly converted to zero, biasing price statistics and free share.
- Platform fields are missing for 84.6% of observations and genre fields for 9.6%.
- Ranks reflect scraped listing order, not exact sales; sales volume, revenue, reviews, ratings and verified discounts are not included.

9. Recommendations and Final Conclusion

9.1 Recommendations before submission

- Update price parsing to separately represent explicit free, numeric price and unknown/not displayed. Recheck sample cards, rebuild price_value, and rerun price-related charts.
- Review platform and genre selectors against representative public listing-card HTML and document the revised missingness rates.
- Package the seven raw CSV/JSON snapshots, cleaned CSVs, SQLite database and scraper logs with the notebook so the project can be reproduced and audited.
- Add a table of exact daily summary statistics produced directly from the cleaned dataset.
- Continue to follow robots.txt, request delays, request caps and stop-on-block behavior. Do not bypass HTTP 401, 403 or 429 responses. [1, 2]

9.2 Final Conclusion

The Group 8 itch.io Top Sellers project demonstrates an end-to-end data workflow that connects public-page collection with data cleaning, structured storage and exploratory visualization. According to the supplied notebook's saved outputs, the analysis covers seven daily snapshots from 13 to 19 October 2026, with 500 observations per day and 852 distinct game URLs across the full period. The generated charts describe the observed price distribution, rank-price association, daily listing counts, rank variation, developer appearances, genre frequencies and platform labels.

The rank-price correlation is very weak in the recorded sample (Spearman correlation −0.044). Adventure, Visual Novel, Simulation and Action are the most frequently recorded genre/tag labels. At the same time, the analysis exposes important quality issues: platform labels are missing in 84.6% of records, genre labels are missing in 9.6%, and the current price parser may classify a price that was not displayed as free. As a result, the figures should be presented as preliminary until the price classification is corrected and the complete raw snapshots are included for reproducibility.

Overall, the project's main achievement is demonstrating a traceable process for turning website listings into analyzable data while recognizing the limits of that data. Better handling of missing price values, improved platform extraction and a final rerun against archived raw files will strengthen the reliability of the results and support more defensible conclusions.